

Subject Description Form

Subject Code	EEE521
Subject Title	Tomorrow's Leader in Electric Vehicles
Credit Value	3
Level	5
Pre-requisite/ Corequisite/ Exclusion	Nil
Objectives	This subject provides a comprehensive exploration of the principles, technologies, societal impacts, policy, business, and project aspects of electric vehicles. Through a series of lectures, seminars, workshops, visits, exchanges with professionals, and case studies, students will gain a detailed understanding of the electric vehicles industry, market, and research development. They will also develop critical thinking, analytical, oral, and written communication skills. By the end of the subject, students will be able to appreciate the challenges and opportunities in the electric vehicles field and propose solutions to address them.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the principles, technologies, and research development of electric vehicles. - Understand the industry, societal impacts, and policy of electric vehicles. - Understand the business, market and project aspects of electric vehicles. - Develop critical thinking and analytical skills to address challenges and opportunities in the field of electric vehicles.
Subject Synopsis/ Indicative Syllabus	1. Introduction to electric vehicles 2. Electric vehicle technologies and components 3. Electric vehicle charging and infrastructure 4. Electric vehicle batteries and energy storage systems 5. Electric vehicle motors and drivetrains 6. Electric vehicle market and industry trends 7. Electric vehicle policy and regulation 8. Electric vehicle business models and financing 9. Electric vehicle projects and case studies 10. Societal impacts of electric vehicles The syllabus is indicative and subject to change based on the instructor's discretion and the needs of the course.
Teaching/Learning Methodology	The subject delivery primarily consists of invited lectures from professionals, such as renowned professors, industrialists, and/or institutional/governmental officials. These lectures may also take the form of workshops, seminars, visits, and forums. In addition, tutorials, work examples, case studies, and assignments will be used to complement the lectures and enhance students' learning experience.

	Teaching/Learning Methodology		Intended subject learning outcomes			
			a	b	c	d
	1. Lectures/seminars		✓	✓	✓	✓
	2. Tutorials		✓	✓	✓	✓
	3. Assignment/mini-project/case studies		✓	✓	✓	✓
Assessment Methods in Alignment with Intended Learning Outcomes						
	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Attendance	10%	✓	✓	✓	✓
	2. Assignment	20%	✓	✓	✓	✓
	3. Mini-project	30%				✓
	4. Case studies	40%	✓	✓	✓	
	Total	100%				
	The attendance is to encourage students to actively interact with instructors and participate in class discussion.					
	The assignment is designed to evaluate students' comprehension of subject materials. Students will be required to answer a series of questions for each lecture, seminar, or workshop presented by professionals.					
	The mini-project is intended to provide students with an opportunity to present their critical thinking and analytical skills of a selected topic. An oral presentation and a take-home report are required.					
	The case studies are designed to provide students with an opportunity to present their understanding of subject materials through studying various real-world cases covering technologies, societal impacts, policy and business of electric vehicles. Students will be required to submit a written report for each case study.					
	Student Study Effort Expected	Class contact:				
▪ Lecture		30 Hrs.				
▪ Tutorial and presentation		9 Hrs.				
Other student study efforts:						
▪ Assignment, mini-project and case studies		36 Hrs.				
▪ Self-study		40 Hrs.				
Total student study effort		115 Hrs.				

Reading List and References	1. C.C. Chan and K.T. Chau, “Modern Electric Vehicle Technology,” Oxford University Press, 2001. 2. J. Larminie and J. Lowry, “Electric Vehicle Technology Explained,” Wiley, 2012. 3. K.T. Chau, “Electric Vehicle Machines and Drives – Design, Analysis and Application,” Wiley-IEEE Press, 2015. 4. K.T. Chau, “Energy Systems for Electric and Hybrid Vehicles,” The IET, 2016. 5. T. Muneer, M.L. Kolhe, and A. Doyle, “Electric Vehicles: Prospects and Challenges,” Elsevier, 2017. 6. Z. Ma and S. Zou, “Efficient Auction Games: Theories, Algorithms and Applications in Smart Grids & Electric Vehicle Charging,” Springer, 2020. 7. R. Xiong, “Battery Management Algorithm for Electric Vehicles,” Springer, 2020. 8. D. Kishan, R. Kannan, B.D. Reddy, and P. Prabhakaran, “Power Electronics for Electric Vehicles and Energy Storage: Emerging Technologies and Developments,” CRC Press, 2023. 9. M. Sankir and N.D. Sankir , “Hydrogen Electrical Vehicles (Advances in Hydrogen Production and Storage (AHPS)),” Wiley-Scrivener, 2023. 10. C.C. Chan, G.Y. Zhou, and W. Han, “Integration of Energy, Information, Transportation and Humanity - Renaissance from Digitization,” Elsevier, 2023.
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